NANDHA ENGINEERING COLLEGE

(AUTONOMOUS)

ERODE – 638 052, TAMIL NADU, INDIA.

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

22AIC15 –FULL STACK DEVELOPMENT- PBL - REVIEW 1

YEAR / SEMESTER : III / VI

PROJECT TITLE: URUDHUNAI(SUPPORT) – A PLATFORM THAT CONNECTS PEOPLE WHO NEED HELP WITH PEOPLE NEARBY WHO CAN HELP

Project Details	Reg No. of Candidates	Name of the Candidate	Name of the Supervisor
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SDG: 11 (Sustainable Cities and Communities)	23AI049	PREETHI G	**Designation**
Make Cities And Human Settlements Inclusive, Safe, Resilient And Sustainable.			Assistant Professor

PROBLEM STATEMENT

- **Urban neighborhoods** frequently experience emergencies such as **floods, heavy rainfall, power outages, and heatwaves**. During such events, **residents face difficulty in identifying available local resources** such as **drinking water, generators, shelter, medical assistance, and transport.**
- Currently, communication is managed through:
  - ✓ WhatsApp groups
  - ✓ Social media posts
  - ✓ Phone calls.
- These methods are unstructured, lack verification, and do not provide real-time visibility of nearby resources. There is no dedicated system for hyperlocal resource coordination with privacy protection and trust validation.
- To address these issues, we propose a web-based **platform that connects people who need help with people nearby who can help** that enables structured, real-time, and privacy-aware neighborhood coordination.

ABSTRACT

- **Urudhunai** is a web-based hyperlocal community resource sharing platform designed to improve neighborhood coordination during both normal and emergency situations.
- The system operates in two modes:
  - **Normal Mode**:

    Residents can share tools, temporary resources, and community assistance.
  - **Emergency Mode**:

    Activated by area admin during disasters to prioritize critical resources such as water, shelter, and medical help.

ABSTRACT

Key Features:

- ❖ OTP-based authentication
- ❖ Area code-based segmentation
- ❖ Real-time map view using geolocation
- ❖ Privacy-preserving approximate location display
- ❖ Resource request and approval workflow
- ❖ Trust score calculation
- ❖ Auto-expiry of temporary resources
- ❖ The system enhances micro-level community resilience and structured coordination.

SCOPE OF PROJECT

Urudhunai focuses on developing a hyperlocal community mutual aid platform that operates in both normal and emergency situations.

The scope of the project includes:

- Designing a web-based platform for **neighborhood-level resource sharing**.
- Implementing **OTP-based secure authentication** for user verification.
- Supporting multiple **area codes for segmented community access**.
- Providing **real-time map visualization** of nearby available resources.
- Developing a **dual-mode system** (Normal Mode and Emergency Mode).
- Enabling structured **request and approval workflow** between residents.
- Implementing a **trust score mechanism** to ensure accountability.
- Applying **privacy-preserving location masking** (approximate location display).
- Integrating auto-expiry mechanism for temporary emergency resources.
- Ensuring **scalability to support up to 5,000 users** per neighborhood.

EXISTING TECHNIQUES:LITERATURE SURVEY

SNO	Solution/Reference	Approach	Key Features/Outcome
1	Reuter, C., et al., “Social Media in Crisis Management” (2018)	Use of social media platforms like Twitter and Facebook for emergency communication and information dissemination.	Enables rapid information sharing but lacks structured resource mapping and verification mechanisms.
2	Vieweg, S., et al., “Microblogging During Natural Hazards” (2010)	Analysis of disaster-related tweets to understand public response and coordination.	Provides situational awareness but suffers from misinformation and absence of trust scoring.
3	Meier, P., “Digital Humanitarians” (2015)	Crowd-sourced crisis mapping using volunteer networks and online collaboration tools.	Supports large-scale crisis mapping but not designed for hyperlocal peer-to-peer resource sharing.
4	Community Alert Systems (Government Portals)	Centralized disaster alerts and emergency notifications issued by authorities.	Provides macro-level warnings but lacks neighborhood-level interaction and real-time resource lifecycle control.
5	Neighborhood Sharing Apps (e.g., local community forums)	Local communication platforms for sharing updates and services.	Allows community interaction but does not support emergency mode switching or privacy-preserving geolocation masking.

PROPOSED METHODOLOGY

- The proposed system introduces:
- Area-Based Segmentation
  - Each user belongs to a specific area code.
- Dual-Mode Architecture
  - Normal Mode for everyday community sharing
  - Emergency Mode activated by Area Admin
- Real-Time Map System
  - Displays nearby resources using Mapbox
  - Distance-based filtering.

PROPOSED METHODOLOGY

- Privacy-Preserving Location Model
  - Exact coordinates stored securely
  - Map displays blurred approximate location.
- Request-Approval Workflow
  - User sends request
  - Provider accepts/rejects
- Temporary communication enabled
  - Auto-Expiry Mechanism
- MongoDB TTL index removes expired resources automatically.

SYSTEM ARCHITECTURE

- **Frontend & Client Interface**
  - React.js for dynamic web application development.
  - Mapbox GL JS for real-time interactive resource mapping.
  - Progressive Web App (PWA) support for mobile accessibility.
  - Geolocation API for capturing user coordinates.
  - Redux Toolkit / Zustand for efficient state management.

- **Backend & Server Architecture**
  - Node.js for scalable server-side execution.
  - Express.js for RESTful API development.
  - Socket.io for real-time bidirectional communication.
  - JWT for secure session authentication.
  - Role-Based Access Control (RBAC) for admin management.

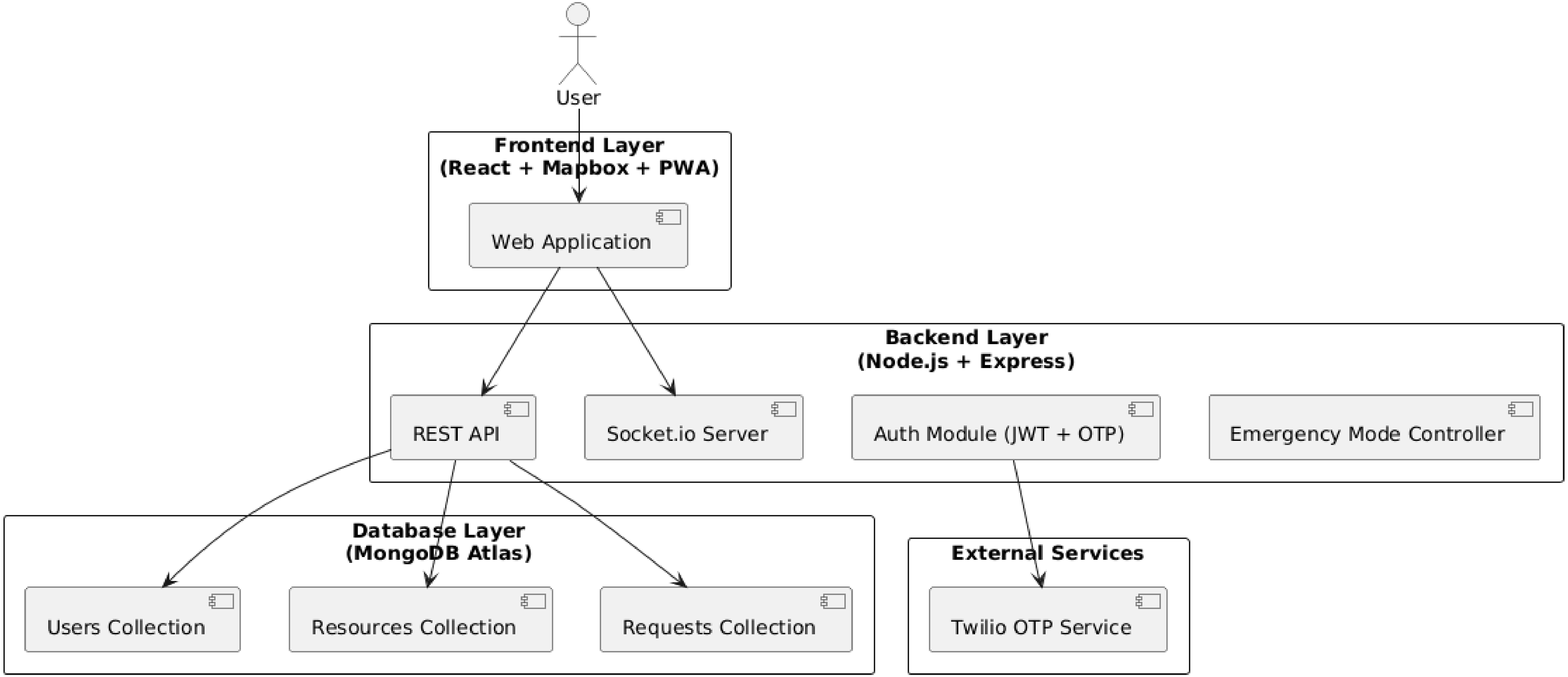
SYSTEM ARCHITECTURE

- **Database & Data Management**
  - MySQLfor SQL storage.
  - GeoJSON indexing for location-based queries.
  - TTL index for automatic resource expiry.
  - Area code-based segmentation for neighborhood isolation.

- **Communication & Networking**
  - Twilio OTP service for phone-based authentication.
  - WebSocket rooms for area-specific real-time updates.
  - REST APIs for frontend ↔ backend ↔ database interaction.

- **Security & Privacy**
  - Approximate location rendering with 50–100m offset.
  - Encrypted JWT tokens for secure access.
  - Trust score algorithm for community reliability tracking.

ARCHITECTURE DIAGRAM



MODULES DESCRIPTION

- **Resource Discovery Module (Map)**
  - Core component that displays nearby community resources on a real-time map
  - Built using **OpenStreetMap with react-leaflet**
  - Shows nearby **Offers (resources)** and **Needs (requests)**
  - Supports category filtering (Medical, Water, Food, Shelter, Power)
  - Calculates distance from the user's location to each resource
  - Protects privacy using **approximate location masking**

  Workflow:

  - ✓User explores map
  - ✓Filters resources by category
  - ✓Views resource details
  - ✓Sends request to the provider

MODULES DESCRIPTION

- **Emergency Governance Module (Admin)**
  - Central administrative dashboard for platform monitoring
  - Displays **active users, system health, and resource statistics**
  - Admin verifies **identity and special volunteer roles**
  - Moderation system for **flagged or fraudulent posts**
  - Maintains **real-time system activity logs**

  Workflow:
  - ✓Users apply for special roles
  - ✓Admin verifies identity and approves roles
  - ✓Admin monitors reports and moderates content

MODULES DESCRIPTION

- **Hyperlocal Alert System**
  - Platform-wide alert broadcasting system
  - Alerts categorized by urgency level
    - Info – General updates
    - Warning – Possible hazards
    - Critical – Immediate danger
    - Resolved – Situation cleared
  - Alerts appear in **community timeline feed**
  - Only **verified admins can publish alerts**
  - Critical alerts appear on **homepage banner**

Workflow

- ✓Admin identifies event
- ✓Creates alert with urgency level
- ✓Alert instantly reaches all users
- ✓Admin marks alert as resolved later

MODULES DESCRIPTION

- **Volunteer & Identity Module**
  - Maintains trusted volunteer database
  - Users register skills such as
    - ✓ First Aid
    - ✓ Electrical
    - ✓ Transport
    - ✓ Shelter management
  - **Trust score system** improves reliability
  - Real-time list of **active volunteers nearby**
  - Different **role permissions** for residents, volunteers, NGOs

  Workflow
    - ✓ User completes profile with skills
    - ✓ Volunteers respond to requests
    - ✓ After task completion users rate each other
    - ✓ Trust score automatically updates

MODULES DESCRIPTION

- ❖ **Emergency Mode**
- ➢ Activates during large-scale emergencies
- ➢ Entire platform switches to **Emergency Dashboard**
- ➢ Key features:
  - ✓ Highlights **critical resources (medical, water, shelter)**
  - ✓ Shows **evacuation and shelter availability**
  - ✓ Simplified high-contrast UI for disaster conditions
  - ✓ Global admin control for activation

  Workflow

  - ✓Admin activates emergency mode
  - ✓System interface switches to emergency state
  - ✓Dashboard prioritizes critical requests
  - ✓Admin disables mode after crisis ends

CONCLUSION

- Resilink provides a structured solution for hyperlocal community coordination.
- The system:
  - Enhances neighborhood resilience
  - Provides real-time visibility of resources
  - Maintains user privacy
  - Encourages reliability through trust scoring
  - Supports emergency escalation through admin control
- It transforms:
  - Unstructured communication → Structured coordination
  - Community isolation → Connected mutual support

FUTURE WORKS

- Implement SMS fallback system and offline-first PWA support to handle low-connectivity disaster scenarios.
- Develop AI-based urgency detection to prioritize critical emergency requests automatically.
- Build a multilingual interface (Tamil & English) to improve accessibility across neighborhoods.
- eate a mobile application using React Native for wider adoption and usability.
- Integrate heatmap analytics and community activity insights for better emergency planning.
- Collaborate with local NGOs and authorities for pilot deployment and real-world validation.

Thank You!